

- Let E be the event that the door is the correct choice
- Let S be the event that the door was switched

$$P(E) = 1/N \quad p(E^c) = (N-1)/N$$

$$P(S|E) = \frac{N-k-1}{N-1} \quad p(S^c|E) = 1 - \frac{N-k-1}{N-1} = \frac{k}{N-1}$$

$$P(S|E^c) =$$

$$1. \quad P(E|S^c) = 1/N$$

$$2. \quad P(E|S) = \frac{1}{N} \cdot \frac{N-1}{N-k-1}$$

3. No matter the number of doors, N, or the number of doors opened after the initial selection, k, Jabba should always switch doors because the probability of winning given that he does switch is higher than the initial probability of selecting correctly the first time (1/N), as it is the initial probability with the number of selections from the leftover doors. This increases his odds of winning.